Password Strength Analyzer

Complete Python source code • Tkinter GUI • Password strength checker and generator

Requirements: Python 3. Tkinter is included with most standard Python installations; no third-party packages are required.

```
import tkinter as tk
from tkinter import messagebox
import re
import string
import secrets

# Check password strength
def check_password():
    password = password_entry.get()

    if not password:
        messagebox.showwarning("Warning", "Please enter a password!")
        return

    score = 0
    suggestions = []

    # Length check
    if len(password) >= 8:
        score += 1
    else:
        suggestions.append("Use at least 8 characters.")

    # Uppercase check
    if re.search(r"[A-Z]", password):
        score += 1
    else:
        suggestions.append("Add uppercase letters.")

    # Lowercase check
    if re.search(r"[a-z]", password):
        score += 1
    else:
        suggestions.append("Add lowercase letters.")

    # Number check
    if re.search(r"[0-9]", password):
        score += 1
    else:
        suggestions.append("Include numbers.")

    # Special character check
    if re.search(r"[^A-Za-z0-9]", password):
        score += 1
    else:
        suggestions.append("Add special characters.")

    # Uniqueness / repetition check
```

Source Code (continued)

```
    if len(set(password)) >= min(6, len(password)):
        score += 1
    else:
        suggestions.append("Avoid repeating characters.")

    # Display result
    if score <= 2:
        strength = "WEAK"
        color = "red"
    elif score <= 4:
        strength = "MEDIUM"
        color = "orange"
    else:
        strength = "STRONG"
        color = "green"

    result_label.config(
        text=f"Password Strength: {strength}",
        fg=color
```

```
    )

    if suggestions:
        suggestion_label.config(
            text="Suggestions:\\n• " + "\\n• ".join(suggestions)
        )
    else:
        suggestion_label.config(
            text="Excellent! Your password meets all checks."
        )


# Generate a strong password
def generate_password():
    characters = (
        string.ascii_letters +
        string.digits +
        string.punctuation
    )

    password = (
        secrets.choice(string.ascii_uppercase) +
        secrets.choice(string.ascii_lowercase) +
        secrets.choice(string.digits) +
        secrets.choice(string.punctuation)
    )

    password += ''.join(
        secrets.choice(characters) for _ in range(12)
```

Source Code (continued)

```
    )

    password = ''.join(secrets.SystemRandom().sample(
        list(password), len(password)
    ))

    password_entry.delete(0, tk.END)
    password_entry.insert(0, password)

    check_password()


# Show / Hide password
def toggle_password():
    if password_entry.cget("show") == "*":
        password_entry.config(show="")
        show_button.config(text="Hide")
    else:
        password_entry.config(show="*")
        show_button.config(text="Show")


# Main window
root = tk.Tk()
root.title("Password Strength Analyzer")
root.geometry("450x450")
root.config(bg="#eaf2f8")
root.resizable(False, False)

# Heading
title = tk.Label(
    root,
    text="Password Strength Analyzer",
    font=("Arial", 18, "bold"),
    bg="#eaf2f8",
    fg="#1a5276"
)
title.pack(pady=25)

# Password input
tk.Label(
    root,
    text="Enter Your Password:",
    font=("Arial", 12),
    bg="#eaf2f8"
).pack(pady=5)

password_entry = tk.Entry(
```

Source Code (continued)

```
    root,
    width=30,
    font=("Arial", 13),
    show="*"
)
password_entry.pack(pady=10)

# Show password button
show_button = tk.Button(
    root,
    text="Show",
    command=toggle_password
)
show_button.pack()

# Check button
tk.Button(
    root,
    text="Check Password",
    command=check_password,
    bg="#2471a3",
    fg="white",
    font=("Arial", 12, "bold"),
    width=20
).pack(pady=15)

# Generate password button
tk.Button(
    root,
    text="Generate Strong Password",
    command=generate_password,
    bg="#239b56",
    fg="white",
    font=("Arial", 11, "bold"),
    width=25
).pack(pady=5)

# Result display
result_label = tk.Label(
    root,
    text="Password Strength: Not Checked",
    font=("Arial", 14, "bold"),
    bg="#eaf2f8"
)
result_label.pack(pady=20)

# Suggestions
suggestion_label = tk.Label(
```

Source Code (continued)

```
    root,
    text="",
    font=("Arial", 10),
    bg="#eaf2f8",
    justify="left",
    wraplength=380
)
suggestion_label.pack(pady=10)

# Run application
root.mainloop()
```